

DAY 2 – POWER SUPPLY SYSTEMS

Course: Diploma in Cyber Security

Topic: Power Supply Systems

Name: HET ACHARYA

What is a Power Supply?

Power supply is a device that provides electric power to different components

What is SMPS?

SMPS stands for **switch mode power supply**, it's convert AC power into DC for computer components

What is UPS?

UPS(**Uninterruptible Power Supply**) when electricity power cuts off then UPS provides limited time backup power its called UPS.

TASK 1 – REAL-WORLD DEVICES

Device using SMPS

1.Desktop PC

In Desktop PC SMPS AC power Converts DC voltages SMPS provides DC to internal components.

2. Laptop – SMPS

A laptop power adapter uses switched-mode power conversion to provide the required DC power to the laptop and charge its battery.

3. Wi-Fi Router – SMPS

A Wi-Fi router uses a switched-mode power adapter to convert mains AC into the low-voltage DC power required by the router.

4. Desktop Computer – UPS

A UPS provides temporary backup power to a desktop computer during a power failure and helps prevent sudden shutdown.

5. CCTV System – UPS

A UPS can keep CCTV equipment running for a limited time when the main power supply fails.

TASK 2 – SMPS INTERNAL COMPONENTS DIAGRAM

A diagram showing the main internal sections of a typical desktop PC SMPS.

PROFESSIONAL POWER SUPPLY UNIT (PSU) FUNCTIONAL DIAGRAM

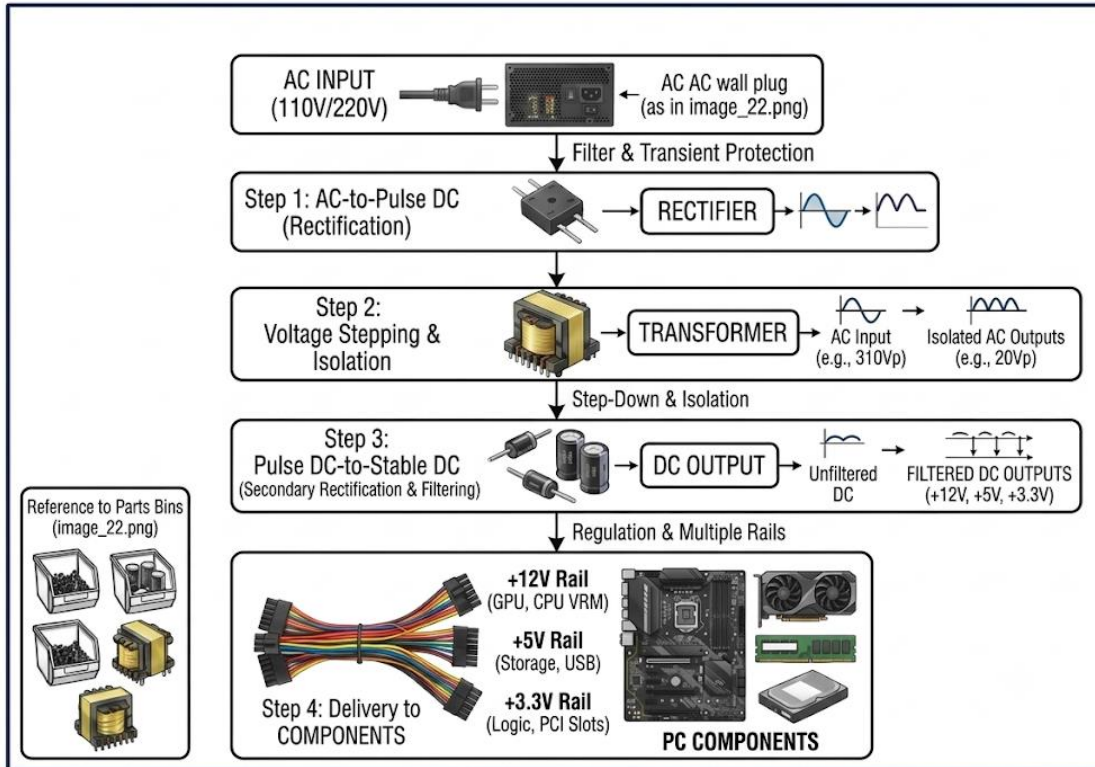


Figure 1: Basic Internal Block Diagram of an SMPS

AC Input:Power enters the SMPS from the AC mains supply.

Rectifier:The rectifier converts AC into DC for the next stage.

Transformer:The transformer changes the voltage to a suitable level for the power supply circuit.

DC Output:The SMPS provides the required DC outputs to PC components.

TASK 3 – POWER CONNECTORS

1. ATX 24-Pin Motherboard Connector

ATX 24-pin connector provides main power connection to motherboard.

2. SATA Power Connector

SATA power connector provides power to SATA SSDs and HDDs.

3. Molex Connector

Molex connector is used to power some older drives, fans and other peripherals .

4. PCIe 6/8-Pin Connector

PCIe 6/8-pin connector provides additional power to graphics cards that require it.

TASK 4 – CONNECTING SMPS IN A GAMING PC

1. Motherboard

Cable: ATX 24-Pin

Connect the 24-pin ATX cable from the SMPS to the main power connector on the motherboard.

2. CPU Power

Cable: CPU/EPS 4/8-Pin

Connect the CPU/EPS power cable from the SMPS to the CPU power connector on the motherboard.

3. Graphics Card

Cable: PCIe 6/8-Pin->If the graphics card requires additional power, connect the correct PCIe power cable from the SMPS to the GPU.

4. Storage Device

Cable: SATA Power Connect the SATA power cable from the SMPS to the SATA SSD or HDD.

Easy Connection Flow

SMPS

|

└─ 24-Pin ATX ──→ Motherboard

|

└─ CPU/EPS ──→ CPU Power

|

└─ PCIe 6/8-Pin ──→ Graphics Card

|

└─ SATA Power ──→ SSD/HDD

CONCLUSION

In this assignment, I learned about **SMPS and UPS systems** and their importance in computer systems.

I learned that SMPS converts AC power into the DC power required by computer components, while UPS provides temporary backup power during a power failure.

I learned that SMPS converts AC power into the DC power required by computer components, while UPS provides temporary backup power during a power failure.

Finally, I learned how to connect the SMPS to the **motherboard, CPU, graphics card and storage devices** using the correct cables.

Finally, I learned how to connect the SMPS to the **motherboard, CPU, graphics card and storage devices** using the correct cables.